# Problem Sheet: Week 11 - solutions

1. This problem is similar to examples seen in class. We first find the probability $q$ that a respondent answers 'Yes'. By the law of total probability we can show that

$$\begin{aligned} q &= \mathbf{P}(\text{'yes'}) \\ &= \mathbf{P}(\{2,3,4,5\}) \cdot \mathbf{P}(\text{'yes'} \mid \{2,3,4,5\}) + \mathbf{P}(\{1,6\}) \cdot \mathbf{P}(\text{'yes'} \mid \{1,6\}) \\ &= \frac{2}{3} \cdot p + \frac{1}{3} \cdot (1-p) \\ &= \frac{p+1}{3}, \end{aligned}$$

since

$$\mathbf{P}(\text{'yes'} \mid \{2,3,4,5\}) = \mathbf{P}(\text{will vote for Party X}) = p$$

$$\mathbf{P}(\text{'yes'} \mid \{1,6\}) = \mathbf{P}(\text{will not vote for Party X}) = 1-p.$$

Note that since $0 \leq p \leq 1$, we must have $\frac{1}{3} \leq q \leq \frac{2}{3}$. As $q$ is a one-to-one function of $p$, by the invariance property of MLEs we have that

$$\hat{p} = 3\hat{q} - 1.$$

Since 110 out of 500 voters answered 'yes', the usual MLE for $q$ is $\frac{11}{50}$. However, this value is below the lower limit of the range for $q$. Hence, we must take $\hat{q} = \frac{1}{3}$ as the MLE for $q$, which corresponds to an MLE for $p$ of $\hat{p} = 0$.

To calculate a 95% confidence interval for $q$, one would usually proceed with the following normal approximation

$$\left( \hat{q} - 1.96\sqrt{\frac{\hat{q}(1-\hat{q})}{500}}, \hat{q} + 1.96\sqrt{\frac{\hat{q}(1-\hat{q})}{500}} \right),$$

and then take the intersection of this interval with the permissible range (1/3, 2/3) to quote the final interval.

However, in this instance, the interval is approximately (0.18, 0.26) and it does not intersect with the permissible range of $q$. So our confidence interval for $q$ is empty and so is the one for $p$! This is because the observed number of 'Yes' responses is considerably lower than one would expect, even if no one actually intended to vote for party X. If this happened in practice, you would suspect that people are not complying with the instructions when they respond to the survey.

2. This problem is similar to examples seen in class. Denote by $X$ the time to relapse in years such that $X \sim \text{Gamma}(2, \lambda)$. We first calculate the probability that the time to relapse is shorter than 1 year as

$$p = \mathbf{P}(X < 1) = \int_0^1 \lambda^2 x e^{-\lambda x} dx = 1 - e^{-\lambda}(1 + \lambda).$$

Note that $p$ is a monotonic (increasing) function of $\lambda$. We know that $\hat{p} = \frac{9}{20} = 0.45$ using the standard MLE for $p$ in the binomial distribution. It follows from the invariance property of MLEs that $\hat{\lambda}$ satisfies

$$e^{-\lambda}(1 + \lambda) = 0.55.$$

Using standard numerical root-finding methods (such as the Newton-Raphson method seen in class) gives $\hat{\lambda} = 1.5234$.

3. To construct the likelihood $L(\lambda, p)$, we need to identify clearly the events that have occurred.

| | |
|---|---|
| Night 1: | 10 meteors are present (as detector A is perfect), |
| | 7 out of the 10 are detected by B. |
| Night 2: | *Unknown* number of meteors is present, |
| | 6 meteors are detected by B. |

Let $A_1$ and $B_1$ be the numbers of meteors detected by A and B on night 1, and let $B_2$ be the number of meteors detected by B on night 2.

From the description we know that $A_1 \sim \text{Poi}(\lambda)$. Conditional on $A_1$, we have that $B_1 \sim \text{Bin}(A_1, p)$. Since the exact number of meteors present on night 2 is unknown, we have that $B_2 \sim \text{Poi}(\lambda p)$ (from examples seen in class).

We can now construct the likelihood $L(\lambda, p)$ as the 'probability of the data given the parameters' for the general case, where $A_1$, $B_1$, and $B_2$ take the values $a_1$, $b_1$, and $b_2$, respectively as

$$\begin{aligned} L(\lambda, p) &\propto e^{-\lambda}\lambda^{a_1} \times p^{b_1}(1-p)^{a_1 - b_1} \times e^{-\lambda p}(\lambda p)^{b_2} \\ &= e^{-\lambda(1+p)}\lambda^{a_1+b_2} \cdot p^{b_1+b_2}(1-p)^{a_1-b_1} \end{aligned}$$

The log-likelihood function is

$$l(\lambda, p) = -\lambda(1 + p) + (a_1 + b_2)\log\lambda + (b_1 + b_2)\log p + (a_1 - b_1)\log(1 - p)$$

Differentiating the log-likelihood function with respect to $\lambda$ and $p$ respectively and equating to zero, we obtain 2 simultaneous equations

$$\frac{\partial}{\partial\lambda} l(\lambda, p) = -(1 + p) + \frac{a_1 + b_2}{\lambda} = 0$$

$$\frac{\partial}{\partial p} l(\lambda, p) = -\lambda + \frac{b_1 + b_2}{p} - \frac{a_1 - b_1}{1 - p} = 0.$$

Solving the two equations gives

$$\hat{p} = \frac{b_1 + b_2}{2a_1 - b_1 + b_2} \quad \text{and} \quad \hat{\lambda} = \frac{2a_1 - b_1 + b_2}{2}.$$

For the data in the question ($a_1 = 10, b_1 = 7$, and $b_2 = 6$), we have

$$\hat{p} = \frac{13}{19} \quad \text{and} \quad \hat{\lambda} = \frac{19}{2}.$$

**Optional:** To show that $(\hat{\lambda}, \hat{p})$ is indeed a maximum point, we need to proceed with the second partial derivative test, which requires showing that

$$\frac{\partial^2}{\partial \lambda^2} l(\hat{\lambda}, \hat{p}) \cdot \frac{\partial^2}{\partial p^2} l(\hat{\lambda}, \hat{p}) - \left( \frac{\partial^2}{\partial p \partial \lambda} l(\hat{\lambda}, \hat{p}) \right)^2 > 0$$

and

$$\frac{\partial^2}{\partial \lambda^2} l(\hat{\lambda}, \hat{p}) < 0 \quad \text{or} \quad \frac{\partial^2}{\partial p^2} l(\hat{\lambda}, \hat{p}) < 0.$$